Digital Twins for Predictive Maintenance in Automotive Aftersales

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***Abstract*—TODO: Abstract here.**

***Index Terms*—Digital Twin, Predictive Maintenance, Industry 4.0, Machine Learning, IoT, Condition Monitoring**

I. Introduction

Today's vehicles are connected by default and continuously transmit operating data [1], [2], so the information needed to judge their technical condition largely exists. Yet aftersales remains reactive: even well-connected vehicles are serviced on fixed time or mileage intervals rather than on the actual condition of their components. Wear parts—brake pads and discs, tyres, ignition parts, or wipers—are replaced only once a fault becomes visible, so customers run them to failure and the unplanned repair often goes to a cheaper third-party garage. For the dealership, which owns the customer relationship, this means losing both the repair and one of only a few yearly customer contacts—each also a chance to advise, retain, and sell [3], [4].

Digital-twin-based predictive maintenance (PdM) can turn this data into foresight. A digital twin (DT) maintains a synchronised virtual representation of the vehicle [5], against which machine-learning models forecast the remaining useful life of components and recommend replacement before failure—keeping the customer at the dealership and enabling transparent, targeted upselling. While the building blocks of DT-based PdM are well studied in industrial and aerospace settings [6], [7], their integration into automotive aftersales remains underexplored.

This paper therefore asks: *what effect does the proactive use of digital twins and predictive maintenance have on the automotive aftersales domain?* Focusing on the German market and working with explicitly stated assumptions, we (i) clarify the interplay of DT and PdM in aftersales, (ii) compare the predictive target state with the interval-based current state to reveal its potential, and (iii) derive recommendations for the dealership. The paper follows the IEEE conference structure: theoretical background, methodology, gap analysis, discussion, and conclusion.

II. Related Work and Theoretical Background

A. Digital Twins

The concept of the digital twin was first introduced by Grieves [5] in a white paper on manufacturing excellence, where it described a virtual replica of a physical product maintained in synchrony with its real-world counterpart throughout the entire asset lifecycle. Originally conceived as a tool for improving production quality in industrial manufacturing, the idea drew on advances in simulation, sensor technology, and computational modelling to propose that a persistent virtual representation of a physical object could serve as the basis for monitoring, analysis, and decision-making. Over the subsequent decade, the concept migrated from its manufacturing origins into aerospace, energy, and infrastructure domains, before emerging as a foundational concept in Industry 4.0 discourse more broadly [6], [8].

In the automotive domain, this general architecture maps directly onto the connected vehicle. The physical entity is the vehicle itself, equipped with an array of on-board sensors and an On-Board Diagnostic (OBD) interface; the virtual representation is a cloud-hosted model that mirrors component states, degradation histories, and operating conditions; and the synchronising link is the over-the-air (OTA) telematics connection that transmits telemetry in near real time [1], [2]. Lopes et al. [1] situate this configuration within a broader software ecosystem in which the digital twin acts as the authoritative source of vehicle state for all downstream services. Architecturally, automotive digital twins are typically realised as modular, microservice-based platforms: a data ingestion layer normalises heterogeneous sensor streams—such as vibration, temperature, voltage, and Controller Area Network (CAN)-bus signals—before forwarding them to a model layer that maintains the current state of each monitored component [9], [10]. The outputs of this stack are then exposed to external consumers such as workshop management systems, Original Equipment Manufacturer (OEM) backends, and customer-facing notification services [1], [7], forming the basis for different predictive maintenance strategies.

B. Vehicle Maintenance Strategies

Vehicle maintenance strategies form a maturity hierarchy: reactive maintenance repairs only after failure [7]; preventive maintenance schedules service at fixed mileage or time intervals, leading to either premature replacement or undetected wear [7], [9]; condition-based maintenance (CBM) couples the service decision to sensor thresholds [9]; and PdM extends CBM by forecasting the future evolution of degradation, estimating the remaining time or distance until end-of-life is reached [7], [11].

PdM relies on three data layers: on-board sensors delivering high-frequency signals such as vibration, current, voltage, and cell temperature [9]–[11]; the OBD interface providing standardised diagnostic codes [2], [12]; and OTA telematics enabling continuous low-latency exchange with the backend [1], [2], [10]. Together, these layers transform the vehicle from a periodically inspected asset into a continuously observed system, a shift first documented in 2009 by Zhang et al. [12].

PdM at this scale is not a stand-alone technique but a service layer built on a DT of the vehicle. As Lopes et al. [1] and Hadraoui et al. [9] describe, the DT supplies the synchronised virtual representation against which sensor streams are interpreted, anomalies detected, and remaining-useful-life estimates computed. Without a DT, telemetry must be processed in isolation; without PdM, the DT remains a passive monitor. In current automotive DT architectures, predictive analytics is therefore typically positioned as a dedicated microservice within the DT's analytics core [1], [7], [10].

C. Machine Learning in PdM

Three problem classes structure machine-learning (ML)-based PdM. Classification assigns sensor patterns to discrete fault categories using Support Vector Machines, Random Forests, or Convolutional Neural Networks on time-frequency representations of vibration signals [7], [11]. Regression targets Remaining Useful Life (RUL) estimation, dominated by Long Short-Term Memory (LSTM) networks for their ability to model long-range temporal dependencies in degradation signals [7], [10]. Anomaly detection operates without explicit labels, learning nominal behaviour and flagging deviations as failure precursors [7], [10]. AbouGrad et al. report classification accuracies of approximately 89% for vibration-based fault detection and 92% for LSTM-based RUL estimation in automotive contexts, though they note that these figures depend strongly on dataset quality and operating conditions.

Data availability is a persistent constraint: AbouGrad et al. and Hadraoui et al. note that preprocessing consumes 70–90% of project time, genuine failure data is structurally scarce, and public benchmarks representative of in-service behaviour are largely absent. Model-based and hybrid approaches that embed physical degradation laws into the learning pipeline are therefore preferred over purely data-driven ones, as they extrapolate more reliably under data scarcity [6], [9], [11]. Beyond accuracy, deployment in safety-relevant contexts raises the question of explainability: deep architectures achieve high accuracy but operate as opaque models whose decisions cannot readily be justified to engineers, customers, or regulators [7]. Explainable AI (XAI) techniques such as feature attribution and attention-based saliency are increasingly integrated into PdM pipelines to expose the reasoning behind a prediction, marking the conceptual bridge from smart to wise information systems [7].

D. Automotive Aftersales

Aftersales encompasses post-purchase activities that help customers access supplier services and resolve product issues through continuous maintenance and upkeep [13]. As a quality-focused product support activity, it creates a sustainable competitive advantage across industries, with automotive being a prime example [14], [15]. In the automotive context, it functions as a dedicated business module bundling maintenance, repair, and warranty offerings to maximise product use throughout the asset lifecycle while meeting financial and customer-performance targets [4], [16]. Building post-sales relationships strengthens customer loyalty, yields actionable insights, and creates upselling and cross-selling opportunities [17], making the automotive aftersales market increasingly attractive as demand for higher-quality service expands [4].

Historically, once the warranty expires, customer engagement with the dealership tends to decline. Within the automotive industry, aftersales services represent a particularly significant source of revenue, with profit margins considerably exceeding those generated through vehicle sales [18]. From an economic standpoint, the aftersales service market has been identified as substantially larger than the primary product sales market across various industries [8], [19], [20]. Retaining these customers requires better information systems and service processes [13]. There is growing recognition that proactively anticipating issues before they arise improves performance, efficiency, and operational excellence [21].

Proactive aftersales improves customer satisfaction and retention by resolving issues before customers encounter them, enabling more productive service interactions [3]. This requires tailoring services to customer needs through tight integration of operational and informational data [13]. ML and DT-based approaches enable predictive interventions and better asset lifecycle optimisation, delivering cost reduction, improved customer satisfaction, and extended asset lifetimes [22].

E. Synthesis and Research Gap

The preceding subsections establish three building blocks: the DT as a synchronised vehicle model, PdM as a prognostic analytics layer built on top of it, and aftersales as the customer-facing business context in which service decisions are communicated and acted upon. Together, they form a potential end-to-end system in which the DT continuously feeds component-state data to ML-based PdM models, whose forecasts are surfaced through aftersales processes to enable proactive service appointments and retain customers at the dealership.

While the referenced contributions establish the technical foundations of DT-based PdM, existing work concentrates either on algorithmic aspects in isolation [7], [11] or on industrial applications [6]. The integration of DT-based PdM into automotive aftersales business models, particularly regarding its implications for customer interaction, remains underexplored. The present work addresses this gap.

III. Methodology

This paper analyses recent academic work on applying DT and PdM in the automotive industry. A structured Literature Review (LR) was conducted to this end. A central element of the LR is to identify, assess, and interpret current research in a structured, transparent, and replicable manner [23]. This minimises selection bias while mapping the current state of the art on DT and PdM in the automotive industry, and serves to identify existing research gaps and future research directions in the field [24].

A gap analysis is conducted as a complementary analytical step. Its central objective is to identify and assess the differences between the current state of automotive aftersales and its potential future development, with particular regard to the combined application of DT and PdM. By systematically comparing current practice with emerging research findings and technological possibilities, the gap analysis highlights areas where significant advancement is yet to be achieved, and provides a basis for deriving concrete implications and future research directions [25].

IV. Gap Analysis

A. Lifecycle Comparison

B. Assumption Framework

Because public, in-service failure datasets representative of this setting are largely absent [7], the comparison is built on an explicit assumption framework rather than on a proprietary dataset. Assumptions are divided into *context* assumptions, which fix the reference case (Table I), *wear-part* assumptions, which size the addressable work (Table II), and *mechanism* assumptions, which govern the transition from the current to the target state (Table III). Each is anchored to a published figure where one exists and flagged as a set value otherwise.

TABLE I
Context assumptions.

ID	Assumption	Value
A1	Reference vehicle	Mid-segment ICE passenger car, German market
A2	Annual mileage	12,300 km/yr [26]
A3	First-owner horizon	8 years [27], [28]
A4	Workshop visits	1.5/yr [29]
A5	Considered wear parts	Brake pads/discs, tyres, wipers, 12 V battery, spark plugs
A6	Service gross margin	50 % (range 45–55 % [28])
A7	Franchised share	43 % [29]

The wear parts in A5 are deliberately limited to components whose wear is gradual, usage-driven and not readily perceived by the driver — the class for which a forecast adds the most decision value and which dominates both between-interval failures and the most common defect categories in the mandatory German vehicle safety inspection (Hauptuntersuchung, HU: brakes, lighting, tyres) [30]. Prescriptive maintenance and powertrain overhauls are out of scope. Typical replacement intervals and representative German market prices (parts plus labour, incl. VAT) are given in Table II; prices are published market ranges [31] and major automotive service providers used at their midpoints.

TABLE II
Wear-part assumptions (midpoints). Prices cover parts plus labour, incl. VAT; midpoints of published German market ranges [31].

Part (A5)	Interval	Events/yr (at A2)	Price
Brake pads (per axle)	40,000 km	0.31	€ 250
Brake discs (+pads, front)	120,000 km	0.10	€ 450
Tyres (set of 4)	45,000 km	0.27	€ 600
Wiper blades (pair)	1 year	1.00	€ 25
12 V battery (fitted)	6 years	0.17	€ 150
Spark plugs (set)	60,000 km	0.21	€ 150

For the mechanism, we model a single lever: the *repair capture rate* c, the share of addressable wear-part work performed at the franchised dealership rather than lost to an independent garage. Its complement $(1 - c)$ is the leakage. The current capture rate, the forecast accuracy, and the proactive-offer conversion are stated in Table III; the target capture rate is not assumed independently but derived from them.

TABLE III
Mechanism assumptions (Current State → Target State). †Conservative lower bound: accuracy improves continuously with data; false-positive forecasts are simply declined offers.

ID	Assumption (Symbol)	Value	Basis
A8	Current capture rate (c_{cur})	0.40	<43 % share [29]
A9	PdM forecast accuracy (a)	0.90	88–92 % LSTM-RUL [7]†
A10	Proactive-offer conversion (k)	0.50	Conserv. estimate; proactive personalised contact outperforms reactive service uptake [3], [15]
—	Target capture rate (c_{tgt}, deriv.)	≈0.67	$c_{\mathrm{cur}} + (1-c_{\mathrm{cur}}) \cdot a \cdot k$

The target capture rate is derived, not assumed: of the leaked share $(1-c_{\mathrm{cur}})$, only the fraction that PdM correctly forecasts (a) and that converts on a proactive offer (k) is recaptured:

$$\begin{aligned} c_{\mathrm{tgt}} &= c_{\mathrm{cur}} + (1 - c_{\mathrm{cur}}) \cdot a \cdot k \\ &= 0.40 + 0.60 \cdot 0.90 \cdot 0.50 \\ &= 0.40 + 0.27 \approx 0.67. \end{aligned} \tag{1}$$

This construction keeps the result honest and never approaches full capture. The conversion rate ($k = 0.50$, A10) is a conservative estimate: proactive, personalised maintenance contact from the customer's own dealership consistently outperforms reactive service approaches in uptake and loyalty [3], [15], yet price competition and the availability of independent garages prevent full conversion. The forecast accuracy (A9) is included for transparency but places no binding constraint on the target state: predictive model performance improves continuously as in-service data accumulates, and a false-positive forecast—recommending a service that is not yet strictly necessary—imposes no cost on the customer, who retains full decision authority and may simply decline the offer.

C. Potential Assessment

The calculation proceeds in three steps: the addressable wear-part value per vehicle, the captured value under each state, and the resulting KPIs at vehicle, fleet and lifecycle level.

Step 1 — Addressable wear-part value: For each part the expected annual value is the number of replacement events per year (Table II) multiplied by its unit value. Summing across A5 yields the annual addressable wear-part service value per vehicle:

$$\begin{aligned} V_{\text{wear}} &= \sum(\text{events/yr} \cdot \text{unit value}) \\ &= 76.9 + 46.1 + 164.0 + 25.0 + 25.0 + 30.7 \\ &\approx €368 \quad \text{per vehicle per year.} \end{aligned} \tag{2}$$

Step 2 — Captured value: Applying the capture rates from Table III:

$$\text{Captured}_{\text{cur}} = c_{\text{cur}} \cdot V_{\text{wear}} = 0.40 \cdot 368 \approx €147 \text{ /veh/yr,} \tag{3}$$

$$\text{Captured}_{\text{tgt}} = c_{\text{tgt}} \cdot V_{\text{wear}} = 0.67 \cdot 368 \approx €247 \text{ /veh/yr.} \tag{4}$$

Applying the service gross margin (A6) gives the captured gross profit.

Step 3 — KPIs: Table IV summarises the comparison. Revenue and gross-profit figures are per vehicle per year; the fleet column scales the gross-profit delta linearly to a representative franchised group of 5,000 active customer vehicles.

TABLE IV
CURRENT-STATE / TARGET-STATE KPI COMPARISON.

KPI	**Current**	**Target**	**Δ**
Repair capture rate	40 %	67 %	+27 pp
Wear-part revenue (/veh/yr)	€ 147	€ 247	+€ 100
Gross profit (/veh/yr)	€ 74	€ 123	+€ 49
Fleet GP, 5,000 veh (/yr)	€ 368k	€ 616k	+€ 248k
Customer touchpoints (/yr)	1.5	∼2.5	+1.0
First-pass inspection rate	baseline	improved	qual.
Lifecycle retention [28]	∼25 %	sustained	qual.

The proactive touchpoint count rises because each high-value forecast event (the wear parts in Table II excluding the wiper case, ≈ 1.0 events/yr) triggers an outreach contact that did not exist in the interval-based current state, roughly doubling the meaningful contact frequency from 1.5 to about 2.5 per year. Each additional contact is also an advisory and upselling opportunity; the parts targeted—brakes and tyres above all—are among the most common defect categories in the German mandatory vehicle safety inspection (HU) [30], so proactive replacement directly improves the customer's first-pass inspection rate, a tangible and communicable benefit.

Sensitivity: The result depends most on the target capture rate. Varying c_{tgt} across a plausible band while holding V_{wear} and the 50 % margin fixed gives a per-vehicle gross-profit delta of:

$$\begin{aligned} c_{\text{tgt}} = 0.60 &\rightarrow \Delta \text{ gross profit} \approx €37 \text{ /veh/yr,} \\ c_{\text{tgt}} = 0.67 &\rightarrow \Delta \text{ gross profit} \approx €50 \text{ /veh/yr,} \\ c_{\text{tgt}} = 0.80 &\rightarrow \Delta \text{ gross profit} \approx €74 \text{ /veh/yr.} \end{aligned}$$

Even at the conservative end the dealership captures additional gross profit on the order of tens of euros per vehicle per year, scaling to a six-figure annual sum across a mid-size fleet—before accounting for the compounding retention effect, namely that dealer-serviced customers are substantially more likely to purchase their next vehicle from the same dealership [32]. The economic case is therefore robust to the assumptions rather than an artefact of any single optimistic value.

D. Limitations

The analysis is intentionally assumption-driven and carries corresponding limitations. First, the figures are illustrative: V_{wear}, the capture rates and the conversion factor are anchored to published market and accuracy data but are not fitted to an operational dataset, and no prototype was built; the contribution is a transparent model of the potential, not a measured outcome. Second, the PdM accuracy assumption (A9) is retained for completeness but places no binding constraint on the target state. Model accuracy improves continuously as in-service data accumulates [7], and a false-positive forecast—predicting wear that has not yet fully materialised—imposes no cost: the customer retains full decision authority and may simply decline the proactive offer. A real deployment would still see accuracy vary by component and data quality, but neither concern fundamentally limits the approach. Third, and most consequential for practice, the target state presumes the dealership can actually obtain the vehicle's telemetry. Under the EU Data Act (applicable from 12 September 2025) the data subject is the vehicle user, who may authorise a third party such as the dealership to receive in-vehicle data on fair, reasonable and non-discriminatory terms, while the OEM remains the data holder; personal data additionally requires a lawful basis under the General Data Protection Regulation (GDPR) [33]. Access is thus contingent on customer consent, OEM provisioning in a usable format, and business-to-business (B2B) compensation — a non-technical adoption barrier that the model does not price in. Finally, two scope caveats apply: tyres, though included as a wear part, are often served by specialist channels and may leak independently of the dealership's PdM capability; and the shift to electric vehicles will erode some ICE-specific wear-part volume (spark plugs, exhaust), shifting the addressable basket toward brakes, tyres and the high-voltage battery over the fleet's lifetime.

V. Discussion and Recommendations

- Interpretation der Ergebnisse
- Implikationen für Forschung und Praxis

VI. Conclusion & Future Work

- Zusammenfassung der wichtigsten Erkenntnisse
- Beitrag der Arbeit
- Ausblick auf weitere Forschung

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Erklärung zur Nutzung von KI-Tools

TODO: KI-Erklärung hier einfügen (gemäß DHBW-Vorgabe).